

water (Farber and Soberman 1956), and a comparison should not be drawn.

(5) Water retention at the time of ovulation has been observed by Thorn et al. (1938), Geiringer (1951), and Greene and Dalton (1953). It is important to know if seizures are common at this time; but this is not yet known, the difficulty of determining the precise time of ovulation being an obvious handicap. We have not yet found an epileptic with predominantly ovulatory epilepsy; and, so far as we could detect, none of our patients experienced the ovulatory tension state (*Mittelwahn* of Geiringer 1951).

(6) We have not had the opportunity of investigating either healthy people, or epileptics with a severe degree of water retention in whom seizures would be more likely if oedema caused them. The cases described by Thomas (1933, case 2) and McCullagh (1941), however, had the features of increased intracranial pressure but no seizures. That menstrual oedema may affect the brain, therefore, cannot be denied but seems to be very rare.

(7) The therapeutic aspects of our conclusions are of interest because it has been suggested that, since acetazolamide promotes diuresis and increases the output of sodium it should be used in menstrual epilepsy to lessen the amount of sodium and water retained; for the same reason it has been used in the premenstrual syndrome (Greenblatt 1955). In 2 epileptic and 1 healthy women we observed this effect when acetazolamide was given premenstrually and it was maintained for three days. But with continued administration the menstrual retention of sodium was not necessarily altered, and the over-all decay curve and the level of total exchangeable sodium were not disturbed. The effect on the patients' seizures could not be assessed, but from clinical experience acetazolamide is known to benefit some epileptics when given premenstrually (Ansell and Clarke 1956).

If our present conclusions about the relationship of seizures to the menses are correct, therefore, acetazolamide cannot be acting entirely by affecting the premenstrual retention of salt and must have another mode of action. Recent work (Millichap et al. 1955, Millichap 1956) indicates that its anticonvulsant action is probably on carbonic anhydrase of the brain, and our findings give some support to this.

If water retention is not directly responsible for the premenstrual and menstrual exacerbation of epileptic seizures, it seems possible that the complex hormonal changes that produce it (Thorn et al. 1938, Gillman 1942, Preedy and Aitken 1956) may play a prominent part.

Summary

The findings in a few female epileptics support the contention of earlier workers that seizures are more common just before and during the menstrual period.

Some of the possible aetiological factors are discussed.

By studying the body-weight, the premenstrual retention of water, and the total body-water, it was found that a disturbance of water storage was not primarily responsible.

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CATAMENIAL EPILEPSY

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SINCE Gowers (1881) and others observed a menstrual exacerbation of fits, it seems to have been tacitly accepted that in some women there is some connection between epilepsy and the menstrual cycle. Although reports on comparatively small series have appeared from time to time, the subject aroused little interest until Almqvist (1955), at the end of a long and careful statistical survey, denied the existence of catamenial epilepsy.

The elucidation of factors influencing the occurrence of seizures is basic to the proper understanding and to the rational management of the epilepsies. To establish facts about one of these factors has more than intrinsic interest. It is only on the basis of fact that hypotheses can be proposed and tested, and from this basis that attempts can be made to advance, however little, towards a solution of the enigma of the epilepsies, which is not so much why people have fits but why they have them when they do.

I report here the main results of an analysis of a very large number of fits (detailed results and discussion of the methods of the analysis will be published elsewhere). These results present very strong statistical evidence of the existence of catamenial epilepsy.

Material

An Edinburgh survey (1954-55 unpublished), in which 73 outpatients recorded 1442 fits in 415 menstrual cycles,

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showed a statistically significant relationship between fits and menstruation. The figures were broken down according to age, frequency and severity of fits, type of fit, presumed aetiology, school record, and present intellectual and affective status. Although the series was not large enough for such a breakdown to be statistically significant, there was a strong suggestion that the catamenial influence which had been demonstrated, though it might be affected by certain of these factors, was not wholly determined by any one or more of them. It was therefore felt that it would be useful to analyse the fits of a group of epileptic inpatients and justifiable to apply the findings from such a special group to the general epileptic population.

At St. Faith's, a hospital for epileptics in Brentwood, Essex, records of both fits and menstruation have been kept for twenty-five years. The records of the 50 presently menstruating women, who had been at the hospital longest, were considered; there was no other selection.

Major fits only were analysed and, owing to the method of recording, 5 or more fits were counted as 5 fits. Minor fits were excluded both because of the difficulty of recording such fits accurately and because of the confusion surrounding the meaning of the term.

Method of Analysis

The first day of menstruation was called *plus 1*, the following day *plus 2*, the day before the onset of bleeding *minus 1*, and so on until 25 cycle-days from *minus 15* to *plus 10* were defined. The number of fits on each of these cycle-days was counted and expressed as a percentage of the expected number of fits per cycle-day obtained by calculation from the average daily number of fits. These percentages for the 25 cycle-days (see table and figure) show the catamenial pattern of incidence of fits.

It is postulated that the physiological changes associated with the menstrual cycle, especially ovulation, influence the occurrence of seizures. In this series there were phases when menstruation was comparatively regular, but there were others which showed an irregularity which was so gross that it was clearly outside the normally wide range of physiological variation. Since it appeared reasonable to assume that there would be a higher proportion of physiological ovular cycles during the regular than the irregular phases, an arbitrary method of selecting regular cycles was defined.

When, in a sequence of three or more cycles, the menstrual type differed by 6 days or less from the mean type for the sequence, the cycles in that sequence were designated as regular cycles. The cycles at the beginning and end of each sequence were excluded, and those remaining were classified as selected regular cycles (S.R.C.s).

The S.R.C.s were then considered separately. The fits per cycle-day were again expressed as a percentage of

CATAMENIAL DISTRIBUTION OF FITS

Cycle-day	All cycles		Selected regular cycles	
	Total fits per cycle-day	Actual (as % of expected) fits per cycle-day	Total fits per cycle-day	Actual (as % of expected) fits per cycle-day
-15	840	94.2	306	91.2
-14	985	110.4	346	103.1
-13	850	95.3	294	87.6
-12	794	89.0	312	93.0
-11	797	89.4	282	84.0
-10	784	87.9	299	89.1
-9	845	94.7	263	78.4
-8	777	87.1	234	69.7
-7	720	80.7	235	70.0
-6	765	85.8	262	78.1
-5	742	83.2	271	80.8
-4	810	90.8	282	84.0
-3	925	103.7	336	100.1
-2	914	102.5	317	94.5
-1	1107	124.1	435	129.6
+1	1246	139.7	488	145.4
+2	1184	132.8	453	135.0
+3	1180	132.3	384	114.4
+4	1036	116.2	353	105.2
+5	1122	125.8	426	126.9
+6	1089	122.1	426	126.9
+7	952	106.7	343	102.2
+8	1076	120.6	377	112.3
+9	980	109.9	338	100.7
+10	991	111.1	340	101.3
Average	891.9 (S.D. 30.1)		335.6 (S.D. 18.0)	

the expected fits (see table), and these percentages were plotted (see figure) to show the catamenial pattern of incidence of fits during phases of comparatively regular menstruation. The pattern for regular cycles was compared with that for the whole series because it was felt that, whereas one could assume that most of the S.R.C.s were physiological, it would not be justifiable to assume that any particular proportion of the other cycles were not.

Results

Fifty patients had 33,468 fits in 939 patient-years. During this time there were 9293 menstrual cycles with 23,511 fits during the 25 cycle-days (-15 to +10). There were 3620 S.R.C.s and 8402 fits during S.R.C. cycle-days.

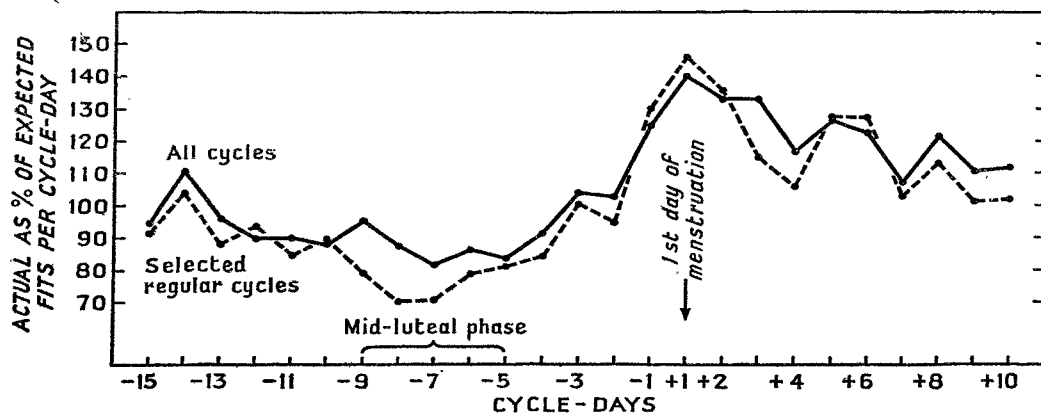
Comparison of the accompanying figure and table shows that, when all cycles are considered, there is an increased incidence of fits immediately before, during, and after menstruation, with a reduced incidence between days -13 and -4 inclusive, and that during phases of comparatively regular menstruation (S.R.C.) the same general pattern is maintained; but that, whereas there is comparatively little alteration in the menstrual exacerbation, there is a significant accentuation of the luteal reduction, particularly during the mid-luteal phase from days -9 to -4 inclusive. Analysis shows not only that nearly all the percentages for individual cycle-days are significant but also that the odds against the comparatively regular form of the graphs being due to chance are enormous.

An analysis of individual catamenial patterns of incidence of fits is not included in this report, but it shows the following facts:

(1) Although 14 patients (28%) showed no significant catamenial influence, the remainder (72%) showed clear evidence of a relationship between the incidence of their fits and the menstrual cycle.

(2) Of those patients showing this latter relationship almost all showed a definite mid-luteal reduction in fits, although the phase of the cycle showing an exacerbation varied, and there might be one or more maxima in any individual graph.

(3) The accentuation of the catamenial



Distribution of fits on cycle-days for whole series (all cycles and selected regular cycles). Fits are expressed as percentage of expected fits per cycle-day.

pattern when S.R.C.s were considered, which was seen in the whole series, was also evident for individual patients.

Discussion

In this analysis no attempt was made to correlate a rhythm of fits with an assumed rhythm of the menstrual cycle. The occurrence of fits was related directly to the first day of menstruation, and a catamenial distribution was demonstrated, despite variations in menstrual pattern. The analysis shows that, in a series of epileptic inpatients menstruation does influence the pattern of fits. In view of the comparable results obtained from unselected outpatients there seems to be no good reason why this influence should not operate in the epileptic population as a whole.

It is usually understood that, if there is such a thing as catamenial epilepsy, there is something about menstruation which increases the proclivity to fits. Contrariwise, the catamenial patterns, especially the mid-luteal reduction in incidence, suggest that there may be some influence during the menstrual cycle which protects a patient from fits; that this influence operates during the luteal phase; and that its withdrawal leads to an irregular exacerbation at a variable time in relation to menstrual bleeding.

Since it is known that, in the non-pregnant woman, progesterone is produced only during the luteal phase and only during ovular cycles, the catamenial pattern and the accentuation of this pattern during phases of regular menstruation could be explained if it were assumed that progesterone exerted a slight but significant anticonvulsant action. The so-called menstrual exacerbation of fits would then be the result of the cessation of progesterone secretion and comparable with that known to occur when anticonvulsants are withdrawn.

Costa and Bonnycastle (1952), who investigated the ability of various steroids to protect dogs from agene-induced convulsions, found that progesterone not only had a considerable anticonvulsant action but also was the most effective of the steroids tested.

Deoxycortone ($-C:O.CH_2OH$) differs from progesterone ($-C:O.CH_3$) only in the introduction of a hydroxy group in its side-chain. Although deoxycortone acetate plays no important part in the treatment of epilepsy, there is no doubt that it may influence epileptic discharges. Woodbury et al. (1950) and Woodbury (1954) report a significant rise in the electric-shock threshold of rats treated with deoxycortone acetate and suggest a relationship to an observed rise in plasma-sodium level. McQuarrie et al. (1942), McQuarrie (1946), and Aird and Gordan (1951) have reported that epileptics have been improved by its exhibition.

This hypothesis does no more than postulate that an observed mid-luteal reduction in the incidence of fits might be due to some primary or secondary effect of the progesterone secreted during this phase. The inter-relationships of the endocrines are so complex that it is hardly likely that the exhibition of progesterone would have a dramatic or sustained effect on the incidence of fits. Nevertheless, if an anticonvulsant action could be proved and its mechanism elucidated, it might prove a useful and physiological adjuvant in therapy.

Further investigations are to be undertaken. Should they, or others, demonstrate this hypothesis to be invalid, the facts which it attempted to explain will remain and stimulate others to offer and substantiate alternative hypotheses.

Summary

The relationship between seizures and menstruation was examined in 50 epileptic inpatients during 939 patient-years.

A method is explained for differentiating between phases of comparatively regular and those of grossly irregular menstruation, and the incidence of fits during the regular phases is estimated.

There is strong statistical evidence that there is a reduction of fits during the luteal phase, with an irregular increase immediately before, during, and after menstruation. The mid-luteal reduction but not the menstrual increase of fits was greater during regular menstruation.

Catamenial epilepsy may be due to an anticonvulsant action of progesterone, with an exacerbation of fits when its secretion stops.

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HYPOPITUITARISM AND GONADAL DYSGENESIS

THE RELATION OF THE PITUITARY GLAND TO SEXUAL DIFFERENTIATION

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It is now clear that women with gonadal dysgenesis ("ovarian agenesis") are most commonly genetically male (Moore et al. 1953, Polani et al. 1954, Carpentier et al. 1955, Grumbach et al. 1955)—assuming that nuclear pattern indicates chromosomal sex. This anomaly may be explained by the work of Jost (1953). He and his co-workers removed, or destroyed by X rays, the foetal gonads at different stages of embryonic development in rabbits. In genetic female embryos their operations were followed by normal or near-normal feminine development; the genitalia became female in type. In male foetuses, if the operations were performed before the genital tract had become differentiated, the foetus developed along fully female lines, and the genitalia became totally feminine. If the operations were done a day or two later, forms of pseudohermaphroditism were produced, with partial feminisation, some development of the prostate, and hypospadias.

Thus it seems that differentiation along normal male lines depends on the presence of functioning testes. Without these, feminisation takes place, whether the foetus was genetically destined to be male or female. Hence we may presume that, in the "male" cases of "ovarian agenesis," the primitive genital streak has not developed in utero into a testis, with the result that the embryo becomes totally feminised while retaining the male nuclear pattern. If the early development of the testes is halted slightly later, partial masculinisation of the genital tract may already have taken place, in which case a "male pseudohermaphrodite" results.

Importance of Pituitary in Sexual Differentiation

In addition to the above-mentioned experiments Jost (1953) reported the results of decapitation of embryos at various stages of development. This is the only known